OZARK

Learning Management System

Modern · Collaborative · Built for Everyone

GROUP 100 · COMP2154

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AGENDA

01

Why Ozark?

The case for a modern LMS

02

Who Is Ozark For?

Target audiences & use cases

03

Why Existing LMS Solutions Fall Short

Gaps in current tools

04

Core Technology

ASP.NET Core · EF Core · SignalR · PostgreSQL

05

Feature Demo

Live walkthrough of key capabilities

06

Conclusion & Value Proposition

What sets OzarkLMS apart

SECTION 01

Why Ozark?

Ozark brings a modern way to manage academic life — cutting through the clutter of legacy tools.

Manage Courses
Create, edit, and organize with ease

Optimize Workflows
Streamline assignments, grading & feedback

Peer Collaboration
Modern student hub & discussion tools

Org Management
Admin controls, enrollments & announcements

Built Different

Unlike legacy LMS platforms that feel like corporate intranets from 2005, OzarkLMS is architected from the ground up for the modern learner.

10+	Core Features
3	User Roles
100%	Responsive Design

Course Management & Collaboration

Full Course Control

Admins and Instructors create any number of courses, manage all content and students, and utilize tools for assignments, modules, and exam creation.

Smarter Than Legacy LMS

While current LMS software has basic conveniences, Ozark's modern data handling improves workflow both visually and functionally — no more clunky interfaces.

Student Hub

Collaboration among peers has never been easier. Modern discussion boards, real-time chat, social profiles, post feeds, and much more.

Grading & Feedback

Instructors view all submissions, assign numeric scores, provide text feedback, and auto-grade quizzes — all from a clean, unified dashboard.

SECTION 02

Who Is Ozark For?

Academic Institutions

Universities, colleges, technical institutes of all sizes

Homeschooling

Structured home education with modern tools

Corporate Training

Onboarding and recruitment training management

Personal Learning

Self-paced individual academic workflows

SECTION 03 · Why Existing LMS Solutions Fall Short

Common Pain Points

- Cluttered, outdated interfaces with poor UX
- No real-time collaboration features
- Fragmented tools + email + drives + portals
- Mobile-unfriendly rigid layouts
- No peer-to-peer social interaction layer

VS

How Ozark Solves This

- Clean Tailwind CSS UI with dark & light mode
- SignalR real-time updates across the platform
- Single unified hub for all academic activities
- Fully responsive — any device, any screen
- Student Hub with social feeds, chat, profiles

SECTION 04

Core Technology Behind Ozark

ASP.NET Core 9 MVC

Server-side framework powering all routing, controllers, and Razor views. Clean MVC separation with full role-based auth via cookie middleware.

Entity Framework + PostgreSQL

EF Core ORM with code-first migrations against a PostgreSQL database. Soft-delete patterns, query filters, and split queries for complex data.

SignalR Real-Time

VoteHub broadcasts live post votes, new comments, and chat updates. Front-end JS connects once and receives push events without page reloads.

Tailwind CSS UI

Utility-first styling with full dark/light mode support. Responsive, accessible, and consistent across all views — no Bootstrap bloat.

PDF Toolbox

PdfSharp handles multi-PDF merging. QuestPDF converts .txt files to polished A4 PDFs. In-browser WYSIWYG editor with export support.

Azure Deployment

Hosted on Azure App Service (Canada Central). CI/CD pipeline ensuring the live demo is always up-to-date.

SECTION 05 · Demo of Core Features

Authentication & Roles

Student · Instructor · Admin with cookie auth, profile management, and access control.

Course & Module Management

Full CRUD for courses, modules, module items (files, embeds, pages), and meeting scheduling.

Assignments & Quizzes

File upload, text entry, multi-attempt settings, auto-graded MCQ quizzes, and instructor feedback.

Calendar & Scheduler

Animated calendar view with personal events plus auto-populated assignment due dates.

Student Hub & Chat

Social feed, post voting, threaded comments, group chats, private messaging, and post sharing.

PDF Toolbox

Merge PDFs, convert .txt to PDF, and export from the in-browser rich-text document editor.

Click here for live demo

SECTION 06

Conclusion & Value Proposition

Unified Academic Hub

One platform for courses, assignments, chat, and calendars.

Real-Time Collaboration

SignalR ensures live updates across feeds, chats, and votes.

Built-in Productivity Tools

PDF Toolbox, sticky notes, and a WYSIWYG document editor.

Modern & Responsive

Tailwind UI with dark/light mode — beautiful on any device.

OzarkLMS proves that a focused, well-architected platform transforms administrative friction into streamlined academic success.

Thank You!

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OzarkLMS

A modern, collaborative platform built for the next generation of academic excellence.

Live Demo